

Repository

GitHub: https://github.com/Newterios/educationalLMS_general

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End Term Project — SRE Implementation

EduLMS distributed microservices system with Docker Compose, Docker Swarm, Kubernetes, Terraform, Ansible, Prometheus and Grafana.

Microservices (6)

#	Service	Stack
1	Auth	Go, gRPC
2	Course	Go, gRPC
3	Assessment	Go, gRPC
4	Notification	Go, NATS consumer
5	Payment	Python / Flask
6	User Profile	Python / Flask

SLOs

Service	Availability	p95 latency	Error rate
Auth	≥ 99.5 %	≤ 150 ms	≤ 0.5 %
Course	≥ 99 %	≤ 200 ms	≤ 1 %
Assessment	≥ 99 %	≤ 250 ms	≤ 1 %
Payment	≥ 99 %	≤ 200 ms	≤ 1 %
User Profile	≥ 99 %	≤ 150 ms	≤ 1 %
Gateway	≥ 99.9 %	≤ 50 ms	≤ 0.1 %

Screenshots

1. Containers running

docker ps

2. Microservices responding

microservices

3. Prometheus targets — all UP

targets

4. Prometheus alert rules

alerts

5. Grafana SRE dashboard

grafana

6. Incident simulation — alert in PENDING

incident

Deliverables

- ☒ 6 microservices
- ☒ Docker Compose / Swarm
- ☒ Kubernetes manifests
- ☒ Terraform (AWS + local)
- ☒ Ansible playbooks
- ☒ Prometheus + Grafana + alerts
- ☒ Incident report + postmortem
- ☒ Git link: https://github.com/Newterios/educationalLMS_general